

Roll no: 26k-0006

Ques no 3 (Class Result Processing)

Algorithm:

- 1) Start
- 2) Set total number of students $N=3$.
- 3) For Student = 1 to N
 - a) Initialize Sum-Marks = 0 and Subject-Deficiency-Flag = FALSE
 - b) For Subject = 1 to 5
 - c) Input Marks
 - d) Set Sum-Marks = Sum-Marks + Marks
 - e) IF Marks < 33
Set Subject-Deficiency Flag = TRUEENDFOR
- 4) Calculate Average: $\text{Average} = \text{Sum-Marks} / 5$.
- 5) IF Average ≥ 80
Set Result = "Distinction".
ELSE IF Average ≥ 60 and Average < 80
Set Result = "Pass".
ELSE
Set Result = "Fail".
- 6) IF Subject-Deficiency-Flag == TRUE
Set Result = "fail - subject-Deficiency".
- 7) Display Average and Result.
ENDFOR
- 8) END

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PAC:

Given Data	Processing Required	Output Requirements
• $N = 3$ (No. of students) • 5 subject marks of each student.	• Sum-Marks = sum of 5 Marks • Average = sum-Marks/5. • Check for Distinction (≥ 80), Pass (≥ 60) or fail (< 60). • If any individual marks fail < 33, set override logic flag to swap the category to Fail - Subject Deficiency.	• Total Average per student. • Final Result.

IPO chart:

Input	Processing	Output
• 5 Marks per student	• Use nested loop to collect input scores. • Accumulate marks and compute numerical mathematical average. • On average basis, evaluate base marks tier category. • Check structural failure • Generate outcomes.	• Students numerical average score. • Students outcome text labels.

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Pseudo code:

BEGIN

DECLARE N AS INTEGER = 3, Student-Count,
Subject-Count, Marks, Sum-Marks AS
INTEGER.

DECLARE Average AS REAL

DECLARE Result AS STRING

DECLARE Has-Deficiency AS BOOLEAN

FOR Student-Count = 1 to N DO

Sum-Marks = 0

Has-Deficiency = FALSE

PRINT "Processing Results for student " Student-Count

FOR Subject-Count = 1 to 5 DO

PRINT "Enter Marks of subject " Subject-Count

INPUT Marks

Sum-Marks = Sum-Marks + Marks

IF Marks < 33 THEN

Has-Deficiency = TRUE

ENDIF

ENDFOR

Average = Sum-Marks / 5.0

IF Average ≥ 80 THEN

Result = "Distinction"

ELSEIF Average ≥ 60 THEN

Result = "Pass"

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ELSE

Result = "Fail".

ENDIF

IF Has-Deficiency == TRUE THEN
Result = "Fail- Subject Deficiency"

ENDIF

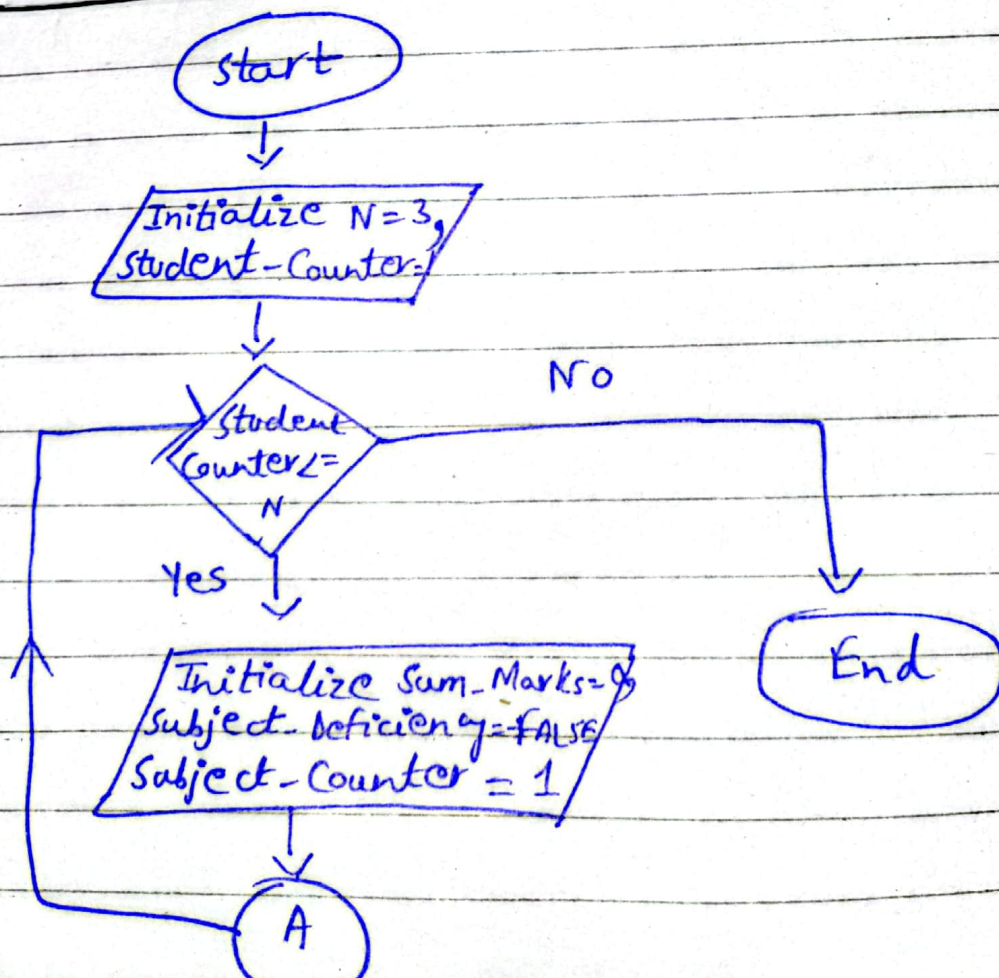
PRINT "Student Average: ", Average

PRINT "Final Classification Result: ", Result.

ENDFOR

END

Flowchart:



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